Policing Spatial Concentration Under LAPD's Post-PredPol DICFP Program

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1 Executive Summary

In April 2020 the Los Angeles Police Department discontinued PredPol, the place-based predictive policing program it had run since 2011, and replaced it with Data-Informed, Community-Focused Policing (DICFP). This study asks whether the spatial concentration of enforcement in Black and Latino neighborhoods changed across that transition, or persisted under a new name. The unit of analysis is the Los Angeles census tract. Enforcement intensity is measured as arrests relative to reported crime, and neighborhood composition as the combined Black and Latino population share from the American Community Survey. The comparison covers 2,362 tract-periods across a PredPol era (2016-2019) and a DICFP era (2022-2024), with 2020-2021 excluded to separate the program change from the pandemic.

Enforcement intensity fell substantially. The citywide median arrest-to-crime ratio dropped from 0.337 to 0.214, and a negative binomial GEE model puts the era effect at an incidence rate ratio of 0.59 ($p < .001$). That decline was not evenly distributed. The interaction of period with tract racial composition is positive and statistically significant (0.172, $p = 0.016$), meaning enforcement fell least where the Black and Latino population share was highest. The racial gradient in enforcement intensity is consequently wider under DICFP than it was under PredPol, rising from 1.79 to 2.13. The program reduced the volume of enforcement without making its distribution across neighborhoods more even.

2 Topic and Research Questions

LAPD ran PredPol, a place-based predictive policing program, from 2011 to 2020 (Mohler et al. 2015). It was discontinued that April, driven by COVID-era budget cuts and by criticism that it concentrated enforcement in already over-policed neighborhoods regardless of underlying crime rates (Lum and Isaac 2016; Mehrotra et al. 2021; Stop LAPD Spying Coalition 2021). LAPD replaced it that same month with Data-Informed, Community-Focused Policing (DICFP), whose stated goals were building trust, reducing violent crime, and assisting victims (Los Angeles Police Department 2020). Optimists read DICFP as a genuine break from PredPol's targeting logic; pessimists read it as the same logic under a new name.

Does LAPD's Data-Informed Community-Focused Policing (DICFP) program show a measurable break from the prior era's spatial concentration of policing activity in Black and Latino neighborhoods, consistent with DICFP's stated goals? Or does enforcement continue along the same spatial lines, past any plausible lagged effect of PredPol itself?

Adjudicating between those two readings starts with the institutional record behind the transition itself. Chief Michel Moore's April 2020 memo attributed PredPol's shutdown specifically to the city's COVID-19 financial crisis and mayoral belt-tightening, not to the racial-bias scrutiny activists and journalists were raising at the same time (Cushing 2020). That the same chief who discontinued PredPol said he still believed in location-based strategies going forward (Cushing 2020) is worth noting before taking DICFP's stated break from PredPol's logic at face value.

Critics argue DICFP borrows its targeting logic from Operation LASER, LAPD's earlier person-based program, and continues to direct enforcement toward areas with already-high call volume (Stop LAPD Spying Coalition 2021; Wille 2021). That claim has been made qualitatively but not tested against tract-level enforcement data.

LAPD's own April 2020 guidebook supports the critics on the narrow question of inputs. Hotspot targeting still runs on Calls for Service, crime and incident reports, and kernel density estimation over 500-1,000 foot "micro-geographic locations" (Los Angeles Police Department 2020, 10–13, 22), functionally PredPol's approach. The community surveys DICFP adds measure trust and satisfaction as outcomes, and nothing in the document describes them feeding back into what counts as a hotspot. That an internal feedback mechanism is undocumented is not evidence that none exists, so this is an evidentiary gap rather than a settled fact.

3 Key Concepts and Variables

Dependent variable: count of arrests per tract-period, modeled as a rate relative to reported crime (see below). Enforcement intensity is the underlying construct. Tracts where arrests exceed what reported crime would predict.

Independent variable: policing-era period. The PredPol era (2016-2019) is compared against the DICFP era (2022-2024), with 2020-2021 dropped to isolate COVID-era confounds from the program change itself. 2015 belongs to the PredPol era but is excluded because four LAPD divisions are missing from that year's crime records while their arrest

records are complete (see Methods and Data Sources).

Unit of analysis: census tract, matched to ACS demographic data. Three groups of tracts are set aside before estimation. Tracts numbered in the 9800s are Census special land use, covering airports, ports, parks, and large institutions; they record incidents but house almost no residents, so a population share computed over a handful of people is not a measurement of neighborhood composition. 20 tract-periods report no crime at all, which leaves the rate undefined. A further 2 have no ACS composition estimate. That leaves 2,362 tract-periods, and every count reported below describes that estimation sample rather than the larger table it is drawn from.

Moderating variable: tract racial composition, the combined share of residents who are non-Hispanic Black or Hispanic/Latino of any race, from ACS table B03002. The two groups are combined because of a limitation in the categories, not because they are interchangeable. Census race and Hispanic origin are asked as separate questions, so "Black" in the tabulated categories means Black alone and not Hispanic, which excludes both Afro-Latino residents and anyone who reports Black together with another race. That exclusion is large: measuring Black residents as Black alone rather than in combination with another race understates the population by roughly 24%, and across the tracts analyzed here 15.9% of residents fall in a "two or more races" category while 23.7% are Hispanic residents reporting "some other race alone." Separating the two groups would therefore assign multiracial residents to one bucket, the other, or neither, leaving the coefficients sensitive to a coding decision with no principled answer. The combined measure is stable under that ambiguity, because a resident who is Black, Latino, or both falls inside it in every coding. What it gives

up is the ability to attribute a change to one group rather than the other, and that cost is real: entered separately, the Black gradient does not change between eras while the Latino gradient widens. That specification is not reported as the headline because its estimates are less trustworthy on the same coding grounds, and because the Black share varies far less across tracts than the Latino share, leaving a null on the Black interaction as consistent with low power as with no effect.

A second and more fundamental mismatch applies to any comparison between arrest records and population data. Census race is self-reported, with Hispanic origin recorded as a separate question from race, and permits multiple selections. LAPD's arrest records assign a single officer-perceived descent code. These do not measure the same construct: a resident who reports Black and Latino on the census appears in one arrest category or the other depending on an officer's judgment. Therefore, any ratio of arrest shares to population shares inherits both measurement systems. This is a general limitation of administrative enforcement data, not a defect of this dataset, but it constrains how finely racial categories can be resolved regardless of which specification is chosen.

4 Hypothesis and Theoretical Rationale

The two readings are directly testable predictions, but they turn on two separate questions: whether enforcement intensity changed at all, and whether its concentration in Black and Latino neighborhoods changed with it. Crossing those questions gives four possible outcomes:

	Gradient narrows	Gradient same or wider
Enforcement falls	1. Genuine reform	2. Volume reform without distributional reform
Enforcement flat	3. Targeted correction without volume change	4. Cosmetic rebrand

Quadrant 1 is the genuine-reform reading and Quadrant 4 the cosmetic-rebrand reading. Quadrants 2 and 3 are not equivalent to either. This project tests both dimensions separately rather than assuming the answer falls on the diagonal.

The bivariate hypothesis is that the change in enforcement intensity between the PredPol and DICFP eras depends on a tract's Black/Latino population share. The independent variable is the interaction of policing-era period with tract racial composition; the dependent variable is the tract's arrest count relative to reported crime. The null is that the two move independently, meaning enforcement intensity changed by the same proportion in every tract regardless of composition.

The theoretical rationale for expecting continuity rather than reform comes from the institutional record above. A program discontinued for budget reasons rather than in response to the bias critique, and rebuilt on the same Calls-for-Service and crime-report inputs feeding the same kernel-density approach to the same micro-geographic grid, has no mechanism by which its spatial distribution would change. Stated goals about trust and community focus operate downstream of where enforcement gets sent, not on the targeting

inputs themselves.

5 Methods and Data Sources

This project tests whether LAPD's shift from PredPol to DICFP corresponds to a measurable change in enforcement intensity, and whether any such change is uniform across neighborhoods or concentrated by racial composition. The analysis compares tract-level arrest-to-crime ratios across the PredPol and DICFP eras, then tests whether the size of that change depends on a tract's Black/Latino population share.

The analysis draws on three public sources. Crime and arrest records come from the Los Angeles Police Department's open data portal, retrieved through its SODA API: the legacy and current crime incident files (Los Angeles Police Department 2024c, 2024d) and the corresponding arrest files (Los Angeles Police Department 2024a, 2024b), covering 2016 through 2024. Tract racial composition comes from American Community Survey five-year estimates, table B03002, using the vintage matched to each era (U.S. Census Bureau 2020, 2024a). Tract boundaries come from the Census Bureau's TIGER/Line shapefiles (U.S. Census Bureau 2024b). Incidents and arrests are geocoded to tracts by spatial join and summed within each era, giving 2,362 tract-periods in the estimation sample.

To test whether the pre/post decline in enforcement intensity is uniform across neighborhoods or concentrated in majority-Black/Latino tracts, period is interacted with tract racial composition. Rather than model the arrest-to-crime ratio directly, arrest counts are modeled as a negative binomial rate with reported crime entered as an offset:

$$\text{Arrests}_{it} \sim \text{NegBinomial}(\mu_{it}, \theta)$$

$$\log(\mu_{it}) = \log(\text{Crime}_{it}) + \beta_0 + \beta_1 \text{Period}_t + \beta_2 \text{PctBlackLatino}_i + \beta_3(\text{Period}_t \times \text{PctBlackLatino}_i)$$

where i indexes census tract and t indexes era. Fixing the coefficient on $\log(\text{Crime}_{it})$ at 1 is what makes this a model of the arrest *rate* relative to crime rather than of counts in isolation, and β_3 is the coefficient of interest. Because the link is logarithmic, a coefficient's exponent is an incidence rate ratio (IRR): the multiplicative change in the arrest rate per one-unit increase in that predictor, above 1 for a higher rate and below 1 for a lower one. Findings reports coefficients this way throughout.

We use negative binomial rather than Poisson, because arrest counts are heavily overdispersed, with a raw variance-to-mean ratio of approximately 1,297. And a population-averaged GEE clustered by tract rather than a GLM with cluster-robust standard errors added afterward, because each tract contributes one observation per era and the two are not independent. All three follow Semenza et al. (2022), who model neighborhood crime counts across Philadelphia census tracts the same way and report that a GEE is more robust to misspecification of the correlation structure than fixed- or random-effects alternatives. The R implementation is included with the submitted analysis code.

6 Findings

Table 1 reports arrest volume and tract racial composition for each era, and Figure 1 shows the citywide crime and arrest series by year. As a preliminary, unconditional check before fitting the negative binomial model described in Methods and Data Sources, the raw arrest-to-crime ratio is then compared citywide across the two eras. That comparison is intended only to establish whether enforcement intensity moved at all, before testing whether the movement varies by tract; it does not test the racial-composition interaction, the question the ratio approach is not suited to answer on its own.

[Table 1 about here]

[Figure 1 about here]

The median arrest-to-crime ratio dropped from 0.337 in the PredPol era to 0.214 in the DICFP era, and the mean dropped from 0.395 to 0.238. The decline is driven by arrests alone. Annualized reported crime is modestly higher in the DICFP era (~223,336/yr to ~223,022/yr), while annualized arrests fall by roughly 40% (~102,720/yr to ~64,126/yr). Enforcement intensity therefore declined while reported crime was flat to rising, rather than tracking any drop in crime itself. A small number of tracts (14 pre, 6 post) had zero recorded crime and were excluded, since the ratio is undefined there.

[Table 2 about here]

[Figure 2 about here]

[Figure 3 about here]

This citywide pattern is consistent with a genuine shift in enforcement intensity between eras, but it says nothing about whether that shift is uniform across neighborhoods. The hypothesis under test is that the change in enforcement intensity between eras depends on a tract's Black/Latino population share; the negative binomial GEE model estimates that dependence directly.

[Table 3 about here]

Two coefficients are estimated precisely. Enforcement intensity fell sharply between eras (IRR 0.59, $p < .001$, a decline of 41% in a tract with no Black or Latino residents), and independently of era, tracts with larger Black and Latino populations sustain higher enforcement relative to reported crime (IRR 1.79, $p < .001$). The first establishes that enforcement practice changed after PredPol ended; the second, that enforcement was unequally distributed across neighborhoods throughout.

The interaction is the term the hypothesis turns on, and it is the smallest and least precisely estimated of the three: 0.172 ($p = 0.016$). It clears conventional statistical significance, but by a far narrower margin than the two main effects, and its substantive magnitude is correspondingly modest. The sign is positive, meaning enforcement intensity fell *less* where the Black and Latino population share was higher. A tract with no Black or Latino residents saw enforcement intensity fall by 41% between eras; an entirely Black and Latino tract by 30%. The reduction reached every kind of neighborhood, but not equally, and the neighborhoods already policed most heavily relative to reported crime were the ones

it reached least. On this evidence the null is rejected: the change in enforcement intensity does depend on tract racial composition. Retaining 2015, excluded for the missing crime records described above, attenuates the interaction to 0.121 ($p = 0.098$), consistent with the inflated PredPol-era intensity in the affected divisions.

[Table 4 about here]

Table 4 shows what those coefficients imply across the composition range, and Figure 4 plots the same relationship. Enforcement intensity falls at every level of composition, but the distance between the ends does not: in the PredPol era a fully Black and Latino tract is arrested at 1.79 times the rate of a tract with no Black or Latino residents, and in the DICFP era at 2.13 times. The disparity is wider after the transition, not smaller.

[Figure 4 about here]

Enforcement fell, but its racial gradient widened rather than narrowed. The result lands in Quadrant 2 of the typology set out in Hypothesis and Theoretical Rationale (volume reform without distributional reform), not Quadrant 1 or Quadrant 4. That placement is interpreted in Discussion.

7 Discussion

7.1 What This Means for the Reform-Versus-Continuity Question

Interestingly, the result lands in Quadrant 2 of the typology: volume reform without distributional reform. Enforcement intensity fell sharply and precisely across the transition, and a change of that size is not what a purely cosmetic rebrand would produce. But the decline was not even. It reached the least Black/Latino tracts fastest and the most Black/Latino tracts least, so the same spatial gradient that motivated PredPol's own bias critique is *wider* under DICFP than it was under PredPol. A program that meaningfully broke from the targeting logic critics objected to should have closed that gap, or at least held it flat.

That combination is closer to what the institutional record in Topic and Research Questions would predict than what either original reading alone would predict. A program cut for budget reasons rather than in response to the bias critique, and rebuilt on the same Calls-for-Service and crime-report inputs feeding the same kernel-density approach to the same micro-geographic grid, would be expected to enforce less simply because there is less capacity to enforce with, without addressing the specific complaint that its predecessor concentrated enforcement in Black and Latino neighborhoods regardless of underlying crime. Nothing in the transition was designed to change that mechanism. This is also the first quantitative test of a claim that has so far been made only qualitatively, by comparing program documents rather than enforcement outcomes.

7.2 Limitations

Several limitations bound what the interaction can mean, and the most consequential is the denominator. Reported crime is not an independent measure of crime: some reports are victim-initiated and roughly independent of patrol intensity, while others originate from officer-initiated contact, which is itself a function of the patrol a tract already receives (Lum and Isaac 2016). Numerator and denominator are generated by the same institution whose behavior the ratio is meant to describe. Reading the widening gradient as a change in enforcement behavior therefore requires that whatever changed in reporting between the eras did not change *differentially by tract racial composition*: the ratio absorbs citywide shifts in reporting without difficulty, but not one concentrated in the same neighborhoods whose gradient is being measured. Even granting that assumption, the result admits three readings the design cannot separate. Enforcement may have contracted least where it was already heaviest; the offense mix recorded in Black and Latino tracts may have shifted toward higher-yield categories; or reported crime may simply have risen faster in whiter tracts. Only the first is a claim about enforcement behavior.

If the assumption fails, the likely direction is known even if the magnitude is not. Victim reporting is lower in over-policed and immigrant communities because of distrust of police (Desmond, Papachristos, and Kirk 2016; Theodore and Habans 2016), and the period boundary coincides with the 2020 protests and a documented national deterioration in police-community relations. Should reporting have declined more in Black and Latino tracts than elsewhere, the denominator would have contracted there specifically and the gradient would widen mechanically, making the estimate an upper bound rather than a measurement.

The convenient assumption that officer-initiated inflation and victim underreporting offset one another is not available: the parallel divergence between hospital and police records of assault injury is race-skewed rather than symmetric (Magee et al. 2021). Nor is there a tract-level remedy, since victimization surveys do not produce estimates below the state, metropolitan, or county level.

Two further limitations are structural rather than measurement-related. Tract racial composition is not held fixed across eras, since each tract's estimate uses the ACS vintage matched to its own period, so the composition coefficient blends how enforcement differs between tracts of different composition with how a single tract's enforcement moves as its own composition changes. Those support different stories about who the widening gradient describes, and a Mundlak-style decomposition entering each tract's panel-mean composition as a separate term would distinguish them. Separately, the design is associative rather than causal. Three developments landed on the same several-month window as the transition, and none can be separated from the program change: the pandemic, which could have moved enforcement in either direction; the budget crisis LAPD itself cites for PredPol's discontinuation (Cushing 2020), which suggests DICFP may have been improvisation rather than considered reform; and the 2020 protests, which plausibly shifted both departmental practice and willingness to report crime. These do not bear symmetrically on the result, since a null would have had to survive all three, whereas the change found here is equally consistent with DICFP, with any one of them alone, or with some combination.

7.3 Future Research

A discretion-based specification would test the same question from an angle that does not depend on the crime denominator at all. In this model, enforcement is compared across offense categories that differ in officer discretion, rather than against reported crime. We leave this for future research rather than attempt it here, since it trades the reporting-bias problem above for a different set of open questions of its own.

7.4 Conclusion

The hypothesis stated in Hypothesis and Theoretical Rationale is supported, though not in the direction the reform reading anticipated. The null, that enforcement intensity changed by the same proportion regardless of tract composition, is rejected: the interaction is positive and statistically distinguishable from zero. But the sign runs the other way from what a genuine break in targeting logic would produce. Enforcement fell everywhere, and fell least where it was already most concentrated relative to reported crime.

That the result lands in a quadrant neither reading anticipated is the study's main surprise. The optimistic and pessimistic readings of DICFP both assumed volume and distribution would move together: that real reform would show up as both less enforcement and a flatter racial gradient, and that a rebrand would show neither. What the data show is one without the other.

Returning to the question posed at the outset, DICFP does not show a measurable break from the prior era's spatial concentration of enforcement in Black and Latino neighborhoods.

It shows less enforcement overall, distributed slightly less evenly than before. That is a limit on the claim rather than a reason to withhold it. The finding is that the observed distribution of enforcement relative to reported crime became less even across racial composition, not that officers changed how they treat individuals of different races. That narrower claim is what the data support, and it is sufficient for the question this project asks, which is whether a program announced as reform reallocated enforcement rather than merely reducing it.

8 Appendices

Table 1: Arrests and tract racial composition by era

Statistic	PredPol Era (Before)	DICFP Era (After)
Mean Arrests / Year	93	50
N Arrests / Year	102,720	64,126
Mean Black/Latino Population (%)	55.7%	54.4%

Note. PredPol era spans 4 years (2016-2019); DICFP era spans 3 years (2022-2024). Per-year figures are

Figure 1: Citywide crime and arrests by year. Crime counts are deduplicated. The 2015 crime figure is shown as published and is understated, since four divisions are missing from that year; 2015 is excluded from the analysis for that reason.

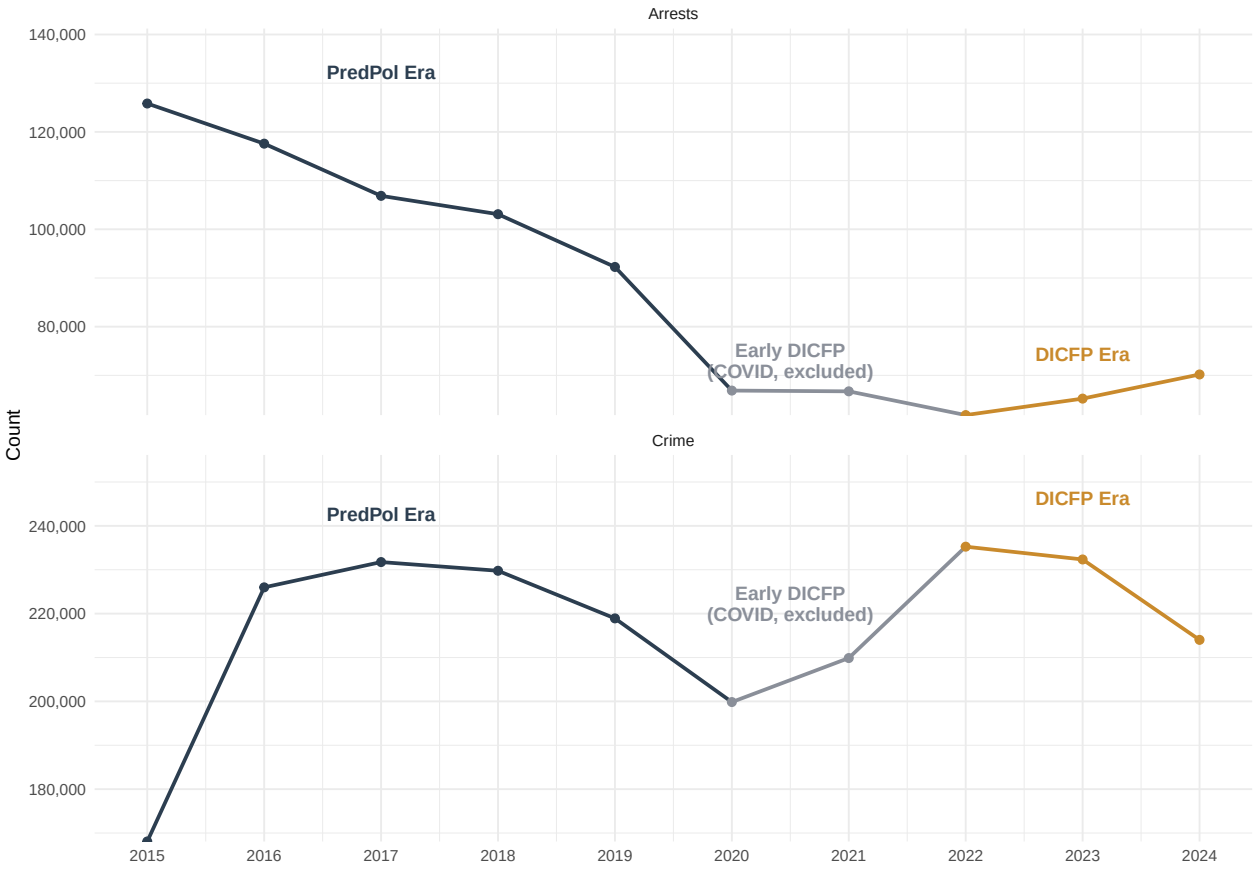


Table 2: Arrest-to-crime ratio by era

Statistic	PredPol Era (Before)	DICFP Era (After)
Mean Ratio	0.395	0.238
Median Ratio	0.337	0.214
Arrests / Year	102,720	64,126
Crimes / Year	223,336	223,022
N Tracts	1,095	1,269
N Dropped	14	6

Note. PredPol era spans 4 years (2016-2019); DICFP era spans 3 years (2022-2024). Per-year figures are

Figure 2: Tract-level arrest-to-crime ratio, PredPol era vs. DICFP era

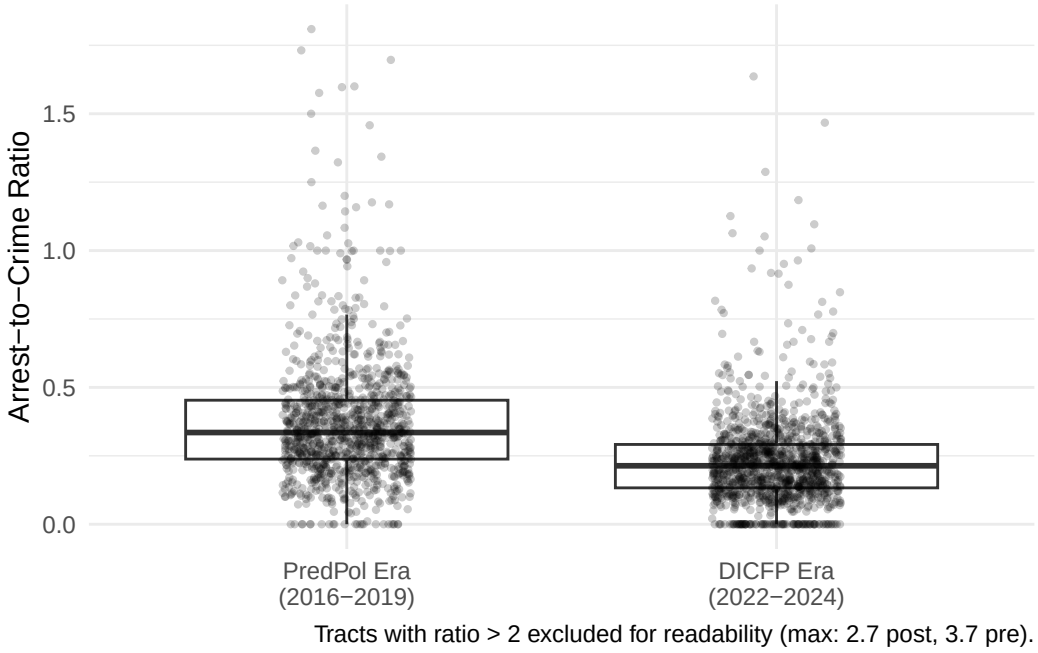


Table 3: Period x racial composition interaction, negative binomial GEE

Term	Estimate	Robust SE	Wald	p	IRR
Intercept	-1.284	0.059	-21.851	0.000	0.277
Period (DICFP Era)	-0.527	0.055	-9.621	0.000	0.590
Black/Latino (%)	0.584	0.077	7.588	0.000	1.794
Period x Black/Latino (%)	0.172	0.072	2.405	0.016	1.188

Table 4: Predicted Arrest-to-Crime Rate by Era and Racial Composition

Black/Latino (%)	PredPol Era (Before)	DICFP Era (After)
0%	0.277	0.163
20%	0.311	0.190
40%	0.350	0.221
60%	0.393	0.257
80%	0.442	0.299
100%	0.497	0.348

Figure 3: Citywide arrest-to-crime ratio by year

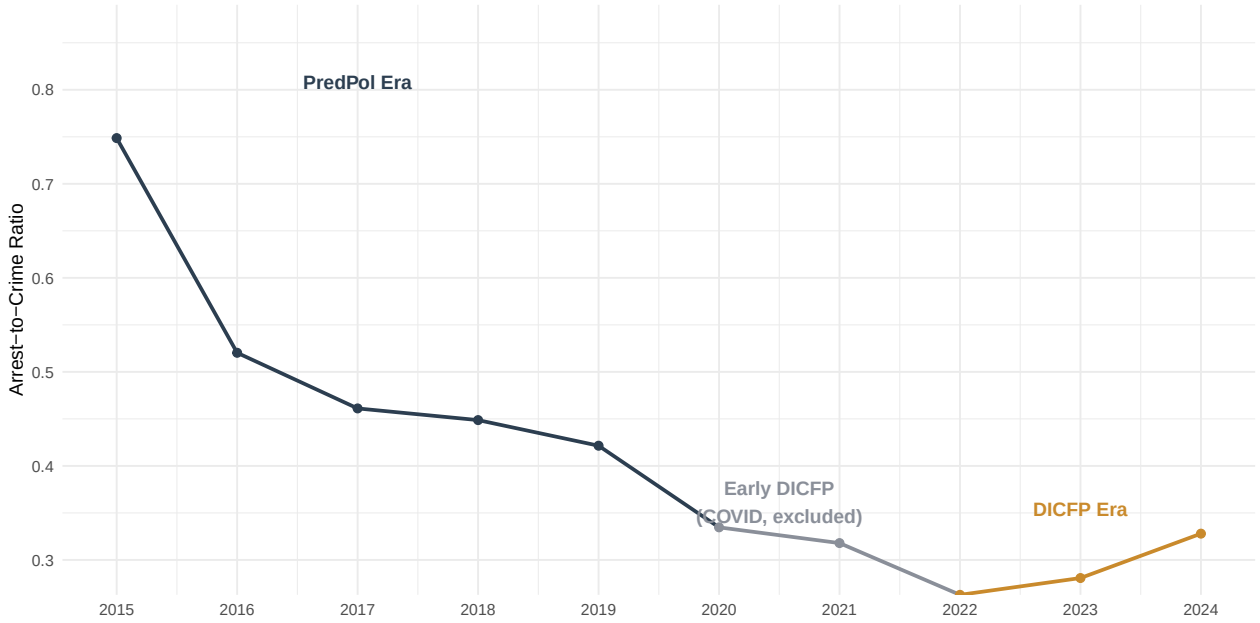
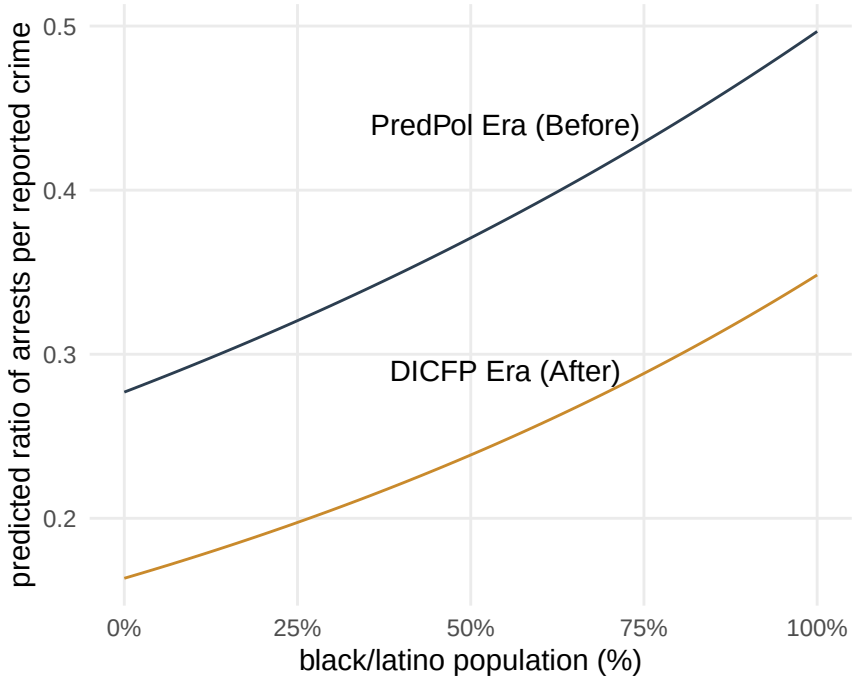


Figure 4: Predicted Arrest-to-Crime Rate by Racial Composition and Era



References

Cushing, Tim. 2020. "LAPD's Failed Predictive Policing Program the Latest COVID-19 Victim." https://www.techdirt.com/2020/04/28/lapds-failed-predictive-policing-program-latest-covid-19-victim/.

Desmond, Matthew, Andrew V. Papachristos, and David S. Kirk. 2016. "Police Violence and Citizen Crime Reporting in the Black Community." *American Sociological Review* 81(5): 857–76.

Los Angeles Police Department. 2020. *Data-Informed, Community-Focused Policing in the Los Angeles Police Department*. Los Angeles Police Department. https://lapdonlinestrgeacc.blob.core.usgovcloudapi.net/lapdonlinemedia/2021/12/data-informed-guidebook-042020.pdf.

———. 2024a. "Arrest Data from 2010 to 2019." https://data.lacity.org/resource/yru6-6re4.

———. 2024b. "Arrest Data from 2020 to Present." https://data.lacity.org/resource/amvf-fr72.

———. 2024c. "Crime Data from 2010 to 2019." https://data.lacity.org/resource/63jg-8b9z.

———. 2024d. "Crime Data from 2020 to Present."

https://data.lacity.org/resource/2nrs-mtv8.

Lum, Kristian, and William Isaac. 2016. "To Predict and Serve?" *Significance* 13(5): 14–19. https://rss.onlinelibrary.wiley.com/doi/full/10.1111/j.1740-9713.2016.00960.x.

Magee, Lauren A. et al. 2021. "Identifying Nonfatal Firearm Assault Incidents Through Linking Police Data and Clinical Records: Cohort Study in Indianapolis, Indiana, 2007–2016." *Preventive Medicine* 149: 106605.

Mehrotra, Dhruv, Surya Mattu, Annie Gilbertson, and Aaron Sankin. 2021. "How We Determined Crime Prediction Software Disproportionately Targeted Low-Income, Black and Latino Neighborhoods." https://themarkup.org/show-your-work/2021/12/02/how-we-determined-crime-prediction-software-disproportionately-targeted-low-income-black-and-latino-neighborhoods.

Mohler, G. O. et al. 2015. "Randomized Controlled Field Trials of Predictive Policing." *Journal of the American Statistical Association* 110(512): 1399–1411. https://www.tandfonline.com/doi/full/10.1080/01621459.2015.1077710.

Semenza, Daniel C., Richard Stansfield, Jessica M. Grosholz, and Nathan W. Link. 2022. "Eviction and Crime: A Neighborhood Analysis in Philadelphia." *Crime & Delinquency* 68(4): 707–32. https://journals.sagepub.com/doi/10.1177/00111287211035989.

Stop LAPD Spying Coalition. 2021. *Automating Banishment: The Surveillance and Policing of Looted Land*. Stop LAPD Spying Coalition. https://automatingbanishment.org.

Theodore, Nik, and Robert Habans. 2016. "Policing Immigrant Communities: Latino Perceptions of Police Involvement in Immigration Enforcement." *Journal of Ethnic and Migration Studies* 42(6): 970–88.

U.S. Census Bureau. 2020. "American Community Survey 5-Year Estimates, Table B03002: Hispanic or Latino Origin by Race."

———. 2024a. "American Community Survey 5-Year Estimates, Table B03002: Hispanic or Latino Origin by Race."

———. 2024b. "TIGER/Line Shapefiles: Census Tracts, Los Angeles County, California."

Wille, Matt. 2021. "LAPD's New Data-Driven Policing Is Just as Racist as Its Old Ways." https://www.inverse.com/input/culture/lapds-new-data-driven-policing-is-just-as-racist-as-its-old-ways.